CONTRACT NOTE REWORK

Data Model

DynamoDB Table & Record Reference

Estimates 1-4

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Overview

This document describes the DynamoDB data model behind the contract note rework. It is a companion to the estimate walkthroughs and the API specification, giving the concrete record shapes, key patterns, and indexes the implementation will use.

The design follows a single-table pattern per bounded area. The bulk of the system, templates, sections, shared sections, rules, versions, data source attachments, and bespoke contracts, lives in one ContractNoteTemplates table, distinguished by partition key (PK) and sort key (SK) patterns. DocuSign envelope tracking sits in its own DocuSignEnvelopes table.

Larger blobs are kept out of DynamoDB: pdf-me section layouts (schema JSON) and rendered PDFs live in S3, referenced from the records by their S3 key.

> ### Reading the key patterns
>
> Values in braces are substituted at runtime, e.g. TEMPLATE#{templateId} becomes TEMPLATE#4f3c... . Records that share a partition key (a template and its sections, versions, rule, and change log) are stored together and retrieved with a single query, which is the core idea behind single-table design.

Tables at a glance

TABLE	ESTIMATES	HOLDS
ContractNoteTemplates	1, 3b, 4	Templates, sections, shared sections, rules, version history, change log, data source attachments, and all bespoke records
DocuSignEnvelopes	2	One record per DocuSign envelope, for status tracking and traceability

Estimate 1 - Templates & Sections

The core template records. Everything owned by a template shares the TEMPLATE#{templateId} partition, so a single query returns the template with its sections, rule, and change log.

Table: `ContractNoteTemplates`

Template

PK `TEMPLATE#{templateId}` SK `METADATA`

ATTRIBUTE	TYPE	DESCRIPTION
`templateId`	*String*	UUID
`name`	*String*	Template display name (unique)
`description`	*String*	Template description
`priority`	*Number*	Evaluation priority (1 = highest)
`sectionCount`	*Number*	Denormalised count of sections
`createdAt`	*String*	ISO 8601 timestamp
`updatedAt`	*String*	ISO 8601 timestamp
`createdBy`	*String*	Cognito username

Section (template-owned)

PK `TEMPLATE#{templateId}` SK `SECTION#{sortOrder}#{sectionId}`

The sort order is baked into the SK so sections return already ordered.

ATTRIBUTE	TYPE	DESCRIPTION
`sectionId`	*String*	UUID
`name`	*String*	Section display name
`sortOrder`	*Number*	Position within the template
`isShared`	*Boolean*	Whether this references a shared section
`sharedSectionId`	*String*	(optional) Reference to a shared section
`schemaS3Key`	*String*	S3 key for the schema JSON
`createdAt`	*String*	ISO 8601 timestamp
`updatedAt`	*String*	ISO 8601 timestamp

Rule

PK `TEMPLATE#{templateId}` SK `RULE`

The specification tree that selects this template (see the Estimate 1 walkthrough).

ATTRIBUTE	TYPE	DESCRIPTION
`specification`	*Map*	JSON specification tree
`updatedAt`	*String*	ISO 8601 timestamp
`updatedBy`	*String*	Cognito username

Template Change Log

PK `TEMPLATE#{templateId}` SK `CHANGELOG#{timestamp}`

ATTRIBUTE	TYPE	DESCRIPTION
`changeType`	*String*	section-added, section-removed, section-reordered, metadata-updated, rule-updated
`description`	*String*	Human-readable description of the change
`createdAt`	*String*	ISO 8601 timestamp
`createdBy`	*String*	Cognito username

Global Secondary Indexes

INDEX	GSI PK	GSI SK	ENABLES
PriorityIndex	`ALL_TEMPLATES (constant)`	`priority (Number)`	Query all templates ordered by priority in a single call

Estimate 1 - Shared Sections & Versions

Reusable sections live under their own partition and are linked to templates by reference records. Section edits are versioned rather than overwritten.

Table: `ContractNoteTemplates`

Shared Section

PK `SHARED_SECTION#{sectionId}` **SK** `METADATA`

ATTRIBUTE	TYPE	DESCRIPTION
`sectionId`	*String*	UUID
`name`	*String*	Section display name
`isTermsAndConditions`	*Boolean*	Whether this is a T&C section
`schemaS3Key`	*String*	S3 key for the schema JSON
`createdAt`	*String*	ISO 8601 timestamp
`updatedAt`	*String*	ISO 8601 timestamp
`createdBy`	*String*	Cognito username

Shared Section Reference

PK `SHARED_SECTION#{sectionId}` **SK** `REF#{templateId}`

One per template using the shared section; how references are tracked and counted.

ATTRIBUTE	TYPE	DESCRIPTION
`templateId`	*String*	Template using this shared section
`templateName`	*String*	Denormalised for display

Section Version

PK `SECTION_VERSION#{sectionId}` SK `VERSION#{timestamp}`

Every schema save appends a version; nothing is overwritten, enabling history and revert.

ATTRIBUTE	TYPE	DESCRIPTION
`versionId`	*String*	UUID
`sectionId`	*String*	Section this version belongs to
`schemaS3Key`	*String*	S3 key for this version's schema JSON
`createdAt`	*String*	ISO 8601 timestamp
`createdBy`	*String*	Cognito username
`description`	*String*	(optional) Change description

Estimate 3b - Data Source Attachments

Data source extensibility adds two record types to the same table. No new table is needed. Attachments hang off the template; dependencies hang off the shared section.

Table: `ContractNoteTemplates`

Template Data Source

PK	SK
`TEMPLATE#{templateId}`	`DATASOURCE#{database}#{tableName}`

Marks a Glue table as attached to a template; its columns become available as fields.

ATTRIBUTE	TYPE	DESCRIPTION
`database`	*String*	Glue database name
`tableName`	*String*	Glue table name
`displayName`	*String*	User-friendly name (defaults to table name)
`attachedAt`	*String*	ISO 8601 timestamp
`attachedBy`	*String*	Cognito username

Shared Section Dependency

PK	SK
`SHARED_SECTION#{sectionId}`	`DATASOURCE_DEP#{database}#{tableName}`

Auto-tracked from the data source fields a shared section uses; drives the missing-dependency prompt.

ATTRIBUTE	TYPE	DESCRIPTION
`database`	*String*	Glue database name
`tableName`	*String*	Glue table name

Estimate 4 - Bespoke Contracts

Bespoke records mirror the template pattern but under a BESPOKE#{bespokeId} partition, keeping one-off documents fully isolated from standard templates. A pending record is written first by the pipeline skip.

Table: `ContractNoteTemplates`

Pending Bespoke Request

PK `BESPOKE_PENDING#{brytNumber}` **SK** `OFFER#{offerReference}`

Written by the render pipeline when it skips a bespoke-flagged customer.

ATTRIBUTE	TYPE	DESCRIPTION
`brytNumber`	*String*	Customer BrytNumber
`offerReference`	*String*	Offer reference from the contract data
`customerName`	*String*	Customer name
`contractDataS3Key`	*String*	S3 key of the original contract JSON
`receivedAt`	*String*	ISO 8601 timestamp when received
`status`	*String*	pending

Bespoke Contract Note

PK	BESPOKE#{bespokeId}

SK	METADATA

ATTRIBUTE	TYPE	DESCRIPTION
bespokeId	*String*	UUID
brytNumber	*String*	Customer BrytNumber
offerReference	*String*	Offer reference
customerName	*String*	Customer name
contractDataS3Key	*String*	S3 key of the contract JSON
clonedFromTemplateId	*String*	(optional) Template ID if cloned
status	*String*	draft, rendering, rendered, failed
currentRenderS3Key	*String*	(optional) S3 key of latest rendered PDF
currentRenderVersion	*Number*	Current render version number
docusignEnvelopeId	*String*	(optional) Envelope ID if sent
docusignStatus	*String*	(optional) sent, completed, declined, expired
createdAt / updatedAt	*String*	ISO 8601 timestamps
createdBy / updatedBy	*String*	Cognito usernames

Bespoke Section

PK `BESPOKE#{bespokeId}` SK `SECTION#{sortOrder}#{sectionId}`

Same shape as a template section; independent copies when cloned from a template.

ATTRIBUTE	TYPE	DESCRIPTION
`sectionId`	*String*	UUID
`name`	*String*	Section display name
`sortOrder`	*Number*	Position within the bespoke contract
`isShared`	*Boolean*	Whether this references a shared section
`sharedSectionId`	*String*	(optional) Reference to a shared section
`schemaS3Key`	*String*	S3 key for the schema JSON
`versionNumber`	*Number*	Current version number
`createdAt / updatedAt`	*String*	ISO 8601 timestamps

Bespoke Render History

PK `BESPOKE#{bespokeId}` SK `RENDER#{version}`

Append-only; one per on-demand render.

ATTRIBUTE	TYPE	DESCRIPTION
`version`	*Number*	Render version number
`pdfS3Key`	*String*	S3 key of the rendered PDF
`renderedAt`	*String*	ISO 8601 timestamp
`renderedBy`	*String*	Cognito username
`status`	*String*	success, failed
`errorMessage`	*String*	(optional) Error details if failed

Estimate 2 - DocuSign Envelopes

DocuSign tracking uses a separate table. One record per envelope, looked up by envelope ID during webhook processing and by Salesforce reference for debugging.

Table: `DocuSignEnvelopes`

Envelope

PK `ENVELOPE#{envelopeId}` SK `METADATA`

ATTRIBUTE	TYPE	DESCRIPTION
`envelopeId`	*String*	DocuSign envelope ID
`salesforceRef`	*String*	Customer Salesforce reference
`contractNoteS3Key`	*String*	S3 key of the original contract note PDF
`customerEmail`	*String*	Email the envelope was sent to
`customerName`	*String*	Customer name on the envelope
`status`	*String*	sent, delivered, completed, declined, expired
`signedPdfS3Key`	*String*	(optional) S3 key of signed PDF once downloaded
`createdAt`	*String*	ISO 8601 timestamp of creation
`updatedAt`	*String*	ISO 8601 timestamp of last status update
`errorMessage`	*String*	(optional) Error / decline reason

Global Secondary Indexes

INDEX	GSI PK	GSI SK	ENABLES
SalesforceRefIndex	`salesforceRef`	`createdAt`	Query all envelopes for a given customer (for debugging)

What lives in S3 (not DynamoDB)

Two kinds of larger data are deliberately kept in S3 and referenced from DynamoDB by key: the pdf-me section layouts (schema JSON, one object per section and per version),

and the rendered and signed PDFs. This keeps DynamoDB items small and cheap to query while letting the bulkier content sit in cost-effective object storage.

- Schema JSON - referenced by schemaS3Key on sections, shared sections, and section versions
- Rendered contract note PDFs - the render pipeline output (Estimate 1)
- Signed PDFs - stored after DocuSign completion (Estimate 2), keyed by Salesforce ref + envelope ID
- Bespoke render PDFs - referenced by pdfS3Key on render history records (Estimate 4)
- Original contract JSON - stored for bespoke customers, referenced by contractDataS3Key

> ### A note on scope
>
> These record shapes come from the estimate design documents and are intended to convey intent and structure. Exact attribute names, optionality, and any additional GSIs may be refined during implementation without changing the overall approach.